

Event-State Theory IX: A Programmatic Bridge Between Variational Classical Physics and Unitary Quantum Interference

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Event-State Theory (EST) proposes a pre-geometric, cost-based explanatory framework for why variational structures arise so broadly in physics. EST does not claim to replace General Relativity (GR) or Quantum Mechanics (QM). Instead, it frames both as effective descriptions emerging from different uses of a single additive, real-valued history cost functional. The classical (GR-compatible) regime is characterized by dominance of extremal-cost histories (minimization), while the quantum regime is characterized by interference between many admissible histories, which requires a unitary representation of the additive cost as a complex phase. This paper is a programmatic bridge: it cleanly separates core postulates from auxiliary assumptions, identifies the precise boundary where purely positive weighting fails to generate interference, and states the minimal conditions under which complex amplitudes of the form $\exp(iS/\hbar)$ are forced. We outline a research program: (I) mathematical sharpening of the cost functional on discrete causal substrates, (II) derivation of continuum variational limits (GR-side), (III) derivation of unitary amplitude propagation (QM-side), and (IV) empirical diagnostics where deviations from standard models may arise.

Author’s Note (Developmental Context).

The papers in the Event-State Theory (EST) series (I–IX) document the progressive development of a discrete causal framework. Early papers introduce core axioms, computational mechanisms, and emergent phenomena, while later papers refine interpretations and identify foundational gaps.

In particular, Papers I–VIII implicitly assume the admissibility of realized event-states. During continued development, this assumption was recognized as nontrivial, motivating separate foundational work addressing the genesis question: how selection and asymmetry arise from a perfect vacuum under the Path of Causal Least Resistance.

This genesis layer is not yet formally integrated into the published EST series and should be regarded as ongoing research. Accordingly, claims in earlier papers are best read as conditional on the existence of an admissible selection process, rather than as a complete account of universe origin.

The present paper (IX) marks a transition point in the project from exploratory construction toward foundational consolidation.

I. SCOPE AND INTENT

EST is descriptive, not prescriptive. The cost functional does not imbue the universe with agency or teleology; it is a structural criterion for which histories are self-consistent under selection. In particular, EST does *not* claim to supersede GR or QM. Its central claim is narrower:

EST provides a pre-geometric, cost-based explanatory framework underlying classical spacetime structure; it is not itself a unification, but it exposes structural conditions under which unification becomes possible.

This paper formalizes the boundary between (I) variational classical structure and (II) quantum interference, showing how both can be organized around one additive cost functional while keeping assumptions explicit.

II. CORE POSTULATES VS AUXILIARY ASSUMPTIONS

We separate what belongs to *EST proper* from what belongs to *conditional extensions*.

A. Core postulates (EST proper)

- Event-state ontology.** The universe is described by a sequence of global states $\{\Psi_n\}_{n \in \mathbb{N}}$ ordered by an update index n . The index n is an order label (not assumed to be physical time).
- Admissible successors.** From Ψ_n there exists a set of admissible successor candidates $\mathcal{C}_n$.
- Real transition cost.** Each candidate transition is assigned a real cost

$$J(\Psi_n \rightarrow \Phi) \in \mathbb{R}, \quad \Phi \in \mathcal{C}_n. \quad (1)$$

4. **History additivity.** For any history $H = (\Psi_0, \dots, \Psi_N)$ define the total cost

$$S[H] := \sum_{n=0}^{N-1} J(\Psi_n \rightarrow \Psi_{n+1}), \quad (2)$$

and require additivity under concatenation $H = H_1 \circ H_2$:

$$S[H_1 \circ H_2] = S[H_1] + S[H_2]. \quad (3)$$

B. Auxiliary assumptions (conditional)

The following are *not* derived from EST; they are optional assumptions that define distinct research branches.

1. **(A1) Positive weighting / entropic selection.** A least-biased assignment of relative weights to histories uses a positive map such as

$$W[H] := \exp\left(-\frac{S[H]}{\lambda}\right), \quad \lambda > 0. \quad (4)$$

2. **(A2) Unitarity / information preservation.** One may additionally require a unitary, composition-compatible representation of history composition (probability conservation at the amplitude level).
3. **(A3) Coarse-grained outcomes.** Observables correspond to coarse-grained sets of fine histories; observation registers outcome statistics, not a selected fine history.

III. WHY POSITIVE WEIGHTS CANNOT PRODUCE INTERFERENCE

The positive weighting (4) is sufficient to rank and suppress histories, and it naturally produces a classical variational backbone. However, because $W[H] > 0$ for all H , sums of such contributions cannot cancel. Therefore purely positive weighting cannot generate destructive interference, a defining feature of QM.

a. Interference threshold. While positive weighting suffices to rank and suppress histories, interference becomes relevant when many admissible histories contribute with comparable total cost. In such regimes, the spread of action values satisfies

$$\Delta S \lesssim \hbar, \quad (5)$$

so that relative phase differences between histories no longer average out under coarse-graining. This marks the boundary between extremal-history dominance (classical behavior) and interference-dominated statistics (quantum behavior), without requiring any change to the underlying cost functional.

This yields the first sharp boundary:

PCLR as cost-minimization and positive weighting explains classical variational structure, but cannot by itself produce interference.

IV. FROM ADDITIVITY TO UNITARY PHASES

To represent histories in a way that allows interference while preserving the additive structure (3), introduce an amplitude map $\mathcal{A}$ from histories to complex numbers,

$$\mathcal{A} : H \mapsto \mathcal{A}[H] \in \mathbb{C}, \quad (6)$$

subject to three structural requirements:

1. **Composition.** $\mathcal{A}[H_1 \circ H_2] = \mathcal{A}[H_1]\mathcal{A}[H_2]$.
2. **Unit modulus.** $|\mathcal{A}[H]| = 1$ for fine-grained histories (unitary, no probability leak at this level).
3. **Continuity.** Small changes in S induce small changes in $\mathcal{A}$.

These requirements force $\mathcal{A}$ to be a continuous unitary character of the additive real line. Therefore,

$$\boxed{\mathcal{A}[H] = \exp(i k S[H])} \quad (7)$$

for some real constant k . Writing $k = 1/\hbar$ introduces the customary action scale $\hbar$:

$$\mathcal{A}[H] = \exp\left(\frac{i}{\hbar} S[H]\right). \quad (8)$$

Interpretive note. This step does not claim to derive unitarity from nothing; it states: *if* one requires a unitary, composition-compatible representation of an additive real cost, *then* the representation must be a complex phase.

V. CLASSICAL VS QUANTUM REGIMES AS TWO USES OF THE SAME COST

From the same real additive functional $S[H]$, two effective regimes appear:

A. Classical/variational dominance (GR-compatible skeleton)

When the contribution of non-extremal histories is suppressed (e.g., because the effective phase varies rapidly or because weighting concentrates mass near minima), the dynamics is dominated by extremal histories:

$$H_{\text{dom}} \approx \arg \min_H S[H]. \quad (9)$$

This yields discrete Euler–Lagrange-type consistency conditions and, in continuum limits, standard variational equations (geodesics and field equations on emergent geometry).

B. Quantum/interference dominance

When many admissible histories contribute comparably, outcomes depend on interference:

$$\mathcal{A}_\alpha := \sum_{H \in \mathcal{H}_\alpha} \exp\left(\frac{i}{\hbar} S[H]\right), \quad P(\alpha) := |\mathcal{A}_\alpha|^2, \quad (10)$$

where $\mathcal{H}_\alpha$ is the set of fine-grained histories compatible with a coarse-grained outcome α . In this regime, phases (not merely costs) control observed statistics.

VI. WHAT THIS PAPER CLAIMS AND DOES NOT CLAIM

A. Claims

1. An additive real history cost provides a unified *structural* language for variational classical physics.
2. Positive history weighting cannot yield interference.
3. Under explicit unitarity and composition requirements, the unique continuous mapping from additive cost to unit-modulus contributions is a complex phase $\exp(iS/\hbar)$.
4. Classical and quantum behaviors correspond to different effective uses of the same $S[H]$: minimization vs summation with phase.

B. Non-claims

1. **No claim** that EST replaces GR or QM.
2. **No claim** that unitarity is derived from EST alone.

3. **No claim** that this paper completes a theory of everything.
4. **No claim** that any particular microscopic form of J is uniquely fixed here.
5. **No claim** of simulated reality. References to 'computation,' 'cost,' or 'optimization' are structural analogies, not ontological claims. EST does not posit that the universe is a computer, but that its behaviors are well-described by selection principles formally analogous to computational processes.

VII. RESEARCH PROGRAM ROADMAP

This program invites participation along clearly separable lines:

1. **Mathematical foundation.** Specify $J(\Psi_n \rightarrow \Psi_{n+1})$ on discrete causal substrates; define admissibility $\mathcal{C}_n$; prove well-posedness and invariance properties.
2. **Classical emergence.** Derive discrete extremal conditions and continuum limits; identify when an effective metric description arises and how Einstein-like variational structure appears.
3. **Quantum emergence.** Formalize amplitude assignment (7) and derive effective propagation equations under coarse-graining; characterize when interference dominates and how Born-type statistics arise.
4. **Diagnostics and falsifiability.** Identify regimes where discrete substrate effects or low causal multiplicity could produce deviations from standard GR/QM predictions, and specify measurable signatures.

VIII. CONCLUSION

EST is positioned here as a disciplined explanatory framework: it clarifies why least-resistance/variational structure is ubiquitous, and it identifies the precise additional condition (unitary representation of additive cost) required for quantum interference. In this sense, EST does not unify GR and QM at the level of equations; it unifies them at the level of structural necessity, conditional on explicit assumptions. This paper is intended as a boundary map and invitation to a falsifiable research program.

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- [9] Wille, T. (2026). *EST VII: Quantum Mechanics as the Statistical Limit of Causal Optimization*. **In preparation.**
- [10] Wille, T. (2026). *EST VIII: Mathematical Derivation of Pre-Causal Resonance*. **In preparation.**
- [11] All code and data are available for independent verification.
Event-State Theory Repository. <https://github.com/SirWillance/Event-State-Theory-EST->